Gym Space Monitoring System

Software Requirements Specification

SRS Version 1.0

Team T08

7 April 2026

Nathan McConnell (Manager)

Ashton Langen

Kevin Rouzaud

Mckenzie Johnston

Julia Marsh

CS 460 Software Engineering

Table Of Contents

1 Introduction **2**

2 General Description **3**

3 Specific Requirements **4**

3.1 Use Cases 4

3.2 External Interfaces 12

3.2.1 Input Events 12

3.2.2 Output Actions 13

3.2.3 Relation to Physical Devices 14

3.3 Control Logic 14

3.3.1 Alert Manager 14

3.3.2 Door Scanner Controller 16

3.3.3 Employee Dashboard 18

3.3.4 Web Service 20

3.3.6 AI Hazard Analyzer 22

3.3.7 Sound Monitor 24

3.3.8 Air Quality/Temp Monitor 24

3.3.9 Machine Monitor 25

4 Design Constraints **26**

4.1 Maintainability 27

4.2 AI and Data Processing Constraints 27

4.3 Environmental Constraints 28

4.4 Privacy Constraints 28

4.5 Equipment and Integration Constraints 28

4.6 System Reliability and Recovery 29

5 Definition of Terms **29**

6 References **31**

1 Introduction

The purpose of this document is to describe the requirements that regulate the design of the Gym Space Monitoring System (GSMS). The system is an automated vision and sensor based solution designed to enhance safety and operation efficiency in a 24-hour gym space. We plan to provide a detailed description of the functional and non-functional requirements for the GSMS, and intend to utilize these specifications as the primary benchmark for future system validation and as a guide for the technical implementation of the software architecture.

The GSMS leverages the benefits of many components such as computer vision and environmental sensors, to transform raw data into actionable insights for gym staff and management. To achieve this, the system needs many inputs granting it the most appropriate and reliable context so that useful actions can be streamed through the outputs. This comprehensive oversight entails AI-powered surveillance cameras across the entire gym space, spanning multiple zones. The cameras will continuously monitor for potential safety hazards, detect falls, and track occupancy levels without requiring constant manual supervision. In addition, passive environmental sensors, including thermostats, smoke detectors, air quality sensors, and certain gym equipment sensors are deployed throughout the facility, ensuring a baseline of environmental safety even within privacy-sensitive locations like the locker rooms where cameras are strictly prohibited. Member engagement and gym space access are streamlined through a front-entrance ID scanner along with a dedicated web application, where gym members can sign up for classes, sign up for a gym membership, and track building capacity to plan their visit, all to enhance the user experience. The sensors on the machines are utilized to track machine specific data, which is then compiled within the system and provided to management, empowering them to make informed, data-driven decisions regarding preventative maintenance and future resource allocation. Not only that but alerts are also directed at the manager, along with the staff and their employee dashboard, where most alerts detected by the system are routed and acted upon by staff per their judgement. The GSMS also comes equipped with a Bulky Buzzer, with the goal of maintaining a comfortable and welcoming atmosphere. The Bulky Buzzer is an automated sound alarm which can be triggered if the gym space is too loud. This alerts members if their noise levels or behaviors become excessively disruptive.

Further details of the system are discussed later in the document, and elaborated on in great detail, providing further context and descriptions of each component of the GSMS. *Section 2* will offer a high-level perspective of the GSMS's functions, user classes, and operating environment, explaining how the system coordinates data from the many sensors and cameras within the system, to achieve real-time safety. *Section 3* defines the use cases as well as the technical inputs and outputs, specifying how the GSMS software interfaces with external hardware, along with details revolving around internal logic that triggers

alerts and other state changes. *Section 4* outlines the boundaries that limit development which we must keep in mind at all stages within the design process. *Section 5* will provide precise definitions for GSMS specific vocabulary, to ensure all readers understand the system's language. *Section 6* contains references used in the document.

2 General Description

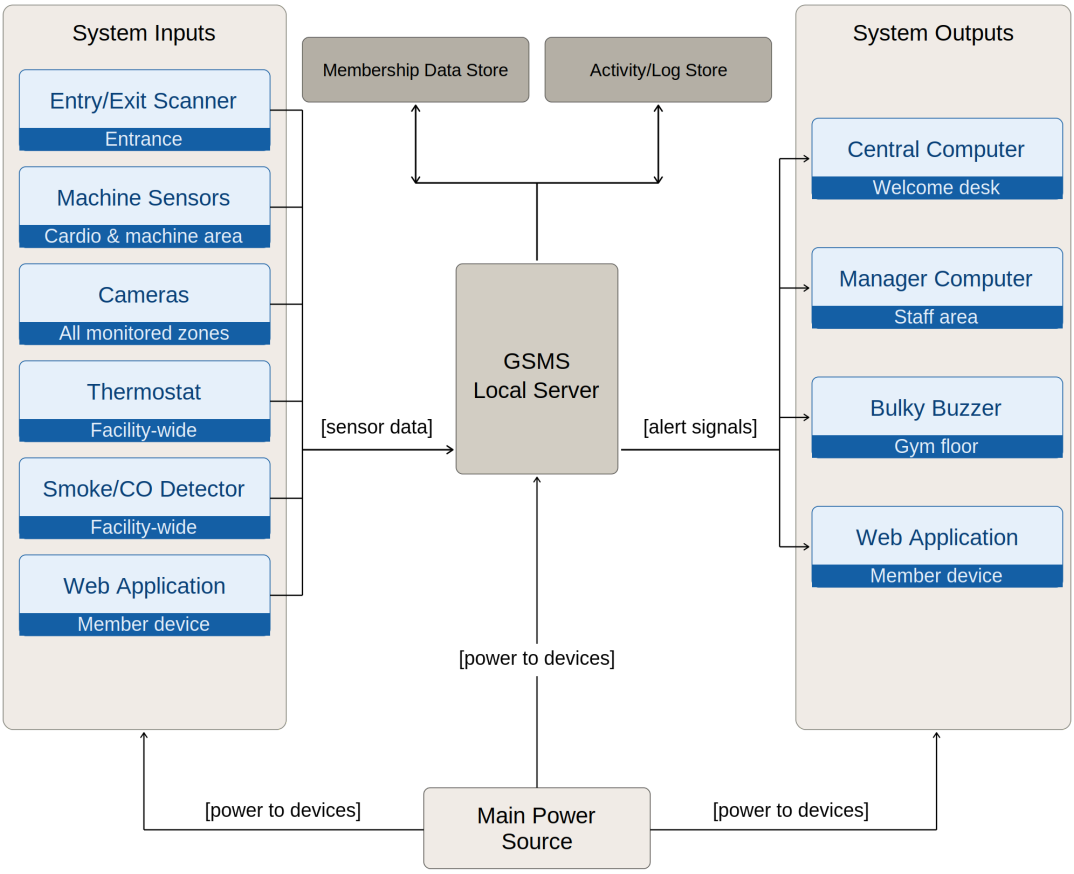

Figure 1: GSMS physical systems block diagram

Figure 1 illustrates the major physical devices connected to the GSMS and shows how they form the foundation that the software operates on. The GSMS is a centralized gym monitoring system made up of interwoven physical devices and software components that work together to improve safety, awareness, and overall gym operations. At a high level, the system gathers information from devices placed throughout the gym, processes that information, and presents the resulting alerts, updates, and reports to the appropriate users.

The physical system includes cameras, exercise machine sensors, thermostats, smoke and carbon monoxide detectors, an entry/exit scanner, staff and manager computers, the Bulky Buzzer, and a web application used by members. These components are distributed throughout the gym space based on their purpose. Cameras and sensors monitor activity and environmental conditions, the scanner tracks member access, and the web application supports membership and class-related interactions. Together, these devices provide the software with the real-world information needed to monitor the facility.

The software coordinates the data collected from these devices and supports the system's major functions. Information gathered from the physical environment is used to detect unsafe conditions, track gym occupancy, monitor equipment usage, and support staff. The GSMS then communicates relevant information back to staff and management through alerts and logs sent to employee dashboards, while selected member-facing information is reflected through the web application. The alerts can be classified as a general alert, a warning alert, or a critical alert, distinguishing themselves based on the event which triggered them. The general alerts are for logging only, while the warning and critical alerts require a validation and resolution from the staff and differentiate themselves on the severity of the event. In some cases, the system may also trigger physical outputs, such as the Bulky Buzzer, when disruptive conditions are detected.

To support unified operation, the GSMS relies on a centralized computing environment that allows the connected components to operate together as a single system. In this way, the physical devices provide the system's inputs, while the software interprets those inputs and helps deliver meaningful outputs that improve safety, response time, and gym management.

3 Specific Requirements

The GSMS is a collection of smaller dynamic objects and components all of which feed a variety of inputs, and provides equally as many varied outputs. The specification for how the GSMS interacts with the gym space through these inputs and outputs is complex and therefore needs to be designed with the utmost care. The particular design along with the methodology will be discussed in further details starting with the first section which will describe all the various interfaces of the system. The details around how the system interacts amongst the various interfaces and how will be discussed. The next section will dive into control logic and control flow, which dictates how the system works, along with helpful diagrams to better illustrate the design.

3.1 Use Cases

The following are definitions of flow between gym members and/or staff with the GSMS, and the expected behavior of the system and how it should react to these interactions. With the focus on the user, little of the GSMS internals are laid out, instead emphasising the expected outputs.

Use Case 1: Sign in to the Gym
Actor: User
Goal: Enter the gym facility
Preconditions:

- User must have an active membership
- User possesses a valid QR code (either physical or digital)
- Entry scanner is operational

Trigger: User scans QR code at entrance
Scenario:

1. User approaches the gym entrance
2. User scans their membership QR code
3. User membership is verified
4. User sees the door unlock
5. User enters the gym

Use Case 2: Responding to a Weight Machine Malfunction
Actor: Staff, Manager
Goal: Resolve malfunctioning weight machine
Preconditions:

- Equipment sensors are operational
- Manager and staff are receiving alerts and messages
- Staff and manager are available to respond

Trigger: Manager receives a malfunction flag from system
Scenario:

1. Manager receives alert indicating machine malfunction
2. Manager acknowledges the alert
3. Staff goes to the machine location
4. Staff inspects the equipment
5. Staff disables or repairs the machine
6. Manager calls for new parts or marks the issue as resolved
7. Manager tracks maintenance logs
8. Manager clears alert and enables machine once it is fixed

Use Case 3: Responding to a Fall
Actor: User, Staff, Manager
Goal: Staff and Manager respond to falls efficiently
Preconditions:
- The cameras are on and collecting data
- The manager and staff are effectively receiving alerts

Trigger: Camera's AI detects state change for longer than 10 seconds.
Scenario:
1. User falls in the gym
2. User is hurt by the fall and stays down
3. Staff and manager receives a critical alert
4. Staff or manager acknowledge the alert
5. Staff or manager assess the user's condition
6. Staff or manager provides assistance or first aid
7. Staff or manager call emergency services if necessary
8. Staff or manager marks alert as resolved

Use Case 4: Dropping a Heavy Free Weight
Actor: User
Goal: Reduce excessive noise in the gym by alerting users to remain below a noise level
Preconditions:
- Sound monitoring system is active in the cameras
- Bulky Buzzer is operational

Trigger: User drops a heavy weight making a loud sound
Scenario:
1. User drop a heavy weight on the ground during lifting, making a loud noise
2. User hears the Bulky Buzzer sound off
3. User realizes they were too loud
4. User uses weights more consciously

Use Case 5: Responding to a Cardio Machine Malfunction
Actor: User, Staff, Manager
Goal: Resolve a cardio machine equipment malfunction
Preconditions:
- Cardio machine sensors are operational
- Staff and managers are receiving alerts

Trigger: Manager receives a malfunction flag from the system
Scenario:
1. Manager receives malfunction alert

2. Manger acknowledges the alert
3. Staff goes to the machine
4. Staff inspects the issue
5. Staff disables or repairs the machine
6. Manager marks the alert as resolved or calls for new parts
7. Manager tracks maintenance logs
8. Manager clears alert and enables machine once it is fixed

Use Case 6: Sign out of the Gym
Actor: User
Goal: Exit the gym facility
Preconditions:
- User has an active membership
- User possesses a valid QR code (either physical or digital)
- Exit scanner is operational

Trigger: User scans QR code at exit
Scenario:
1. User approaches exit doors
2. User scans membership QR code
3. User sees the door unlock
4. User exits the facility

Use Case 7: User drops bar on themself and Staff/Manager Respond
Actor: User, Staff, Manager
Goal: Respond to injured user from weight drop efficiently
Preconditions:
- Cameras and AI object detection are active
- Manager and staff are receiving alerts and messages

Trigger: User drops bar and Staff receive critical alert
Scenario:
1. User drops free weight bar on themselves
2. Staff and manager receive critical alert indicating a user is pinned
3. Staff or manager acknowledge the alert
4. Staff immediately go to the location
5. Staff assists user in removing weight
6. Staff or manager assesses user condition
7. Staff provides first aid or calls emergency services in needed
8. Staff or manager mark alert as resolved

Use Case 8: Responding to Aggression in the Gym
Actor: Staff, Manager
Goal: De-escalate conflict between users
Preconditions:
- Cameras and AI object detection are active
- Staff and manager are receiving alerts and messages

Trigger: Staff and manager receives critical alert
Scenario:
1. Staff and manager receive critical alert indicating aggressive/unsafe behavior
2. Staff or manager acknowledge the alert
3. Staff or manager go to the location
4. Staff or manager intervene to de-escalate the situation
5. Staff or manager ensure all users are safe
6. Staff or manager remove individuals if necessary
7. Staff or manager mark the alert as resolved

Use Case 9: Responding to Overcrowding in a Gym Zone
Actor: Staff
Goal: Respond to unsafe crowding conditions
Preconditions:
- Cameras in the zone are active and sending data
- Cameras and AI object detection model are active
- Manager and staff are receiving alerts and messages

Trigger: AI detects that the number of people in a zone exceeds the allowed threshold
Scenario:
1. Staff receives an overcrowding alert
2. Staff selects the alert and acknowledges the alert using their PIN
3. Staff goes to the affected gym area
4. Staff assesses crowd conditions
5. Staff directs members to other areas or limits access
6. Staff marks the alert as resolved

Use Case 10: Responding to an Obstructed Walkway
Actor: Staff
Goal: Prevent injuries by identifying and responding to tripping hazards
Preconditions:
- Cameras and object detection are monitoring walkways
- Manager and staff are receiving alerts and messages

Trigger: Object is detected as blocking a walkway

Scenario:

1. Staff receives a warning alert for obstruction
2. Staff selects and acknowledges the alert
3. Staff goes to the reported location
4. Staff removes or repositions the obstructing object
5. Staff verifies the walkway is clear
6. Staff marks the alert as resolved

Use Case 11: Responding to Improper Equipment Usage
Actor: Staff
Goal: Reduce injury risk from unsafe usage by responding efficiently
Preconditions:

- Cameras and AI detection are active
- Improper usage patterns are defined
- Manager and staff are receiving alerts and messages

Trigger: AI detects unsafe or improper use of equipment
Scenario:

1. Staff receives a warning alert
2. Staff acknowledges the alert
3. Staff observes the user in question
4. Staff provides guidance or intervention
5. Staff ensures safe usage is restored
6. Staff marks the alert as resolved

Use Case 12: Staff/Manager Acknowledges and Resolves Alert
Actor: Staff, Manager
Goal: Process and resolve system alerts
Preconditions:

- System is actively monitoring and generating alerts
- Staff and Manager has access to central or manager computer
- Alert notification system is operational

Trigger: Staff or Manager receives an alert from the system
Scenario:

1. Staff or Manager receives alert notification
2. Staff or Manager opens alert on dashboard
3. Staff or Manager acknowledges the alert
4. Staff or Manager reviews alert details (e.g. location, type, severity)
5. Staff or Manager goes to the affected location if needed
6. Staff or Manager resolves the issue

7. Staff or Manager marks the alert as resolved in the system

Use Case 13: Manager Reviews Equipment Usage
Actor: Manager
Goal: Make maintenance or purchasing decisions
Preconditions:
- System is collecting equipment usage data
- Manager has access to manager dashboard
- Data storage and reporting system is operational

Trigger: Manager opens dashboard to review equipment data
Scenario:
1. Manager logs into dashboard
2. Manager selects equipment usage reports
3. Usage frequency and duration data is displayed for the user
4. Manager identifies heavily used or underused machines
5. Manager receives maintenance trends
6. Manager decides to schedule maintenance or purchase equipment
7. Manager records or exports report if needed

Use Case 14: Manager Reviews Peak Gym Hours
Actor: Manager
Goal: Optimize staffing and operations
Preconditions:
- Manager has access to dashboard
- Historical data is available

Trigger: Manager request occupancy or usage analytics
Scenario:
1. Manager logs into dashboard
2. Manager selects occupancy or peak hours report
3. Manager identifies peak hours and low-traffice periods
4. Manager adjusts staffing or scheduling accordingly
5. Manager saves or exports report if needed

Use Case 15: User Signs Up for a Class
Actor: User
Goal: Register for a gym class
Preconditions:
- User has an active membership
- User has access to web app or mobile app

- Class schedule is available

Trigger: User selects a class from the schedule

Scenario:

1. User logs into web or mobile app
2. User views available classes
3. User selects a desired class
4. User checks availability
5. User is registered
6. User receives confirmation

Use Case 16: User Attempts to Join a Full Class

Actor: User

Goal: Attempt to register for class

Preconditions:

- User has an active membership
- User has access to web app or mobile app
- Class is at full capacity

Trigger: User selects a full class

Scenario:

1. User logs into app
2. User selects a class
3. User checks availability
4. User sees the class is full and either adds themselves to the waitlist of exits the schedule for that class
5. User selects an alternative or exits

Use Case 17: User Checks Gym Occupancy

Actor: User

Goal: Plan visit based on crowd levels

Preconditions:

- System is collecting occupancy data
- User has access to web app or mobile app

Trigger: User opens occupancy feature

Scenario:

1. User logs into app
2. User selects occupancy view
3. User reviews crowd levels
4. User decides when to visit gym

Use Case 18: Staff/Manager Responds to Environmental Hazard
Actor: Staff, Manager
Goal: Ensure safety during hazardous conditions
Preconditions:

- Environmental sensors (e.g., smoke, temperature, etc.) are active
- System alert system is operational
- Staff and Manager are receiving alerts

Trigger: System detects environmental hazards (e.g., smoke, high temperature, etc.)
Scenario:

1. Staff or Manager receive hazard alert
2. Staff or Manager acknowledge alert
3. Staff or Manager go to affected area
4. Staff or Manager assess severity of hazard
5. Staff or Manager takes appropriate action (e.g., evacuation, equipment shutdown, etc.)
6. Staff or Manager notify emergency services if required
7. Staff or Manager mark alert as resolved once safe

3.2 External Interfaces

The GSMS interacts with users and external devices in order to monitor the gym space and respond appropriately to events taking place within it. These interfaces provide the software with the information it needs from the physical environment and allow the system to communicate useful outputs back to staff, managers, members, and connected devices. Because the GSMS depends on real-world data to function correctly, these external interfaces must be clearly defined so that the system can react to events in a reliable and responsible way. The following subsections describe the major input events, output actions, and their relation to the physical devices used throughout the gym.

3.2.1 Input Events

Input events provide the GSMS with the context it needs to interpret activity within the gym. These events originate from connected devices and user interactions and may result in updates to system state, alerts, logs, or displayed information.

- **Entry/Exit Scanner Event**
  When a gym member presents their QR code to the scanner, a signal is sent to the system to log the member's entry or exit. This allows the GSMS to track current occupancy, total visits, duration of stay, and time-based traffic trends.
- **Exercise Machine Sensor Signal**
  When a gym member uses a monitored machine, the machine sensor sends a signal

to the system indicating activity, duration, or usage count. This data is used to track machine demand, support maintenance planning, and identify high-use equipment.

- **Camera Data Stream**
  Cameras continuously provide visual data from monitored gym areas. This data is evaluated by the GSMS to support functions such as fall detection, aggression detection, overcrowding detection, improper equipment usage detection, and walkway obstruction detection.
- **Thermostat and Air Quality Signal**
  When temperature or air quality values move outside acceptable limits, a signal is sent to the system to indicate that staff may need to investigate the affected area.
- **Smoke/CO Detector Signal**
  When smoke is detected or carbon monoxide rises beyond a safe threshold, the detector sends a signal to the system so that the GSMS can generate a hazard alert for staff or management.
- **Web Application Request**
  When a member interacts with the web application, the system receives requests such as membership registration, class sign-up, or gym occupancy lookup. These requests update the appropriate system records and may change what information is presented back to the user.

3.2.2 Output Actions

After evaluating incoming events, the GSMS may produce one or more output actions to notify users, update records, or activate connected devices.

- **Dashboard Alerts**
  The system sends warning or critical alerts to employee and manager dashboards when unsafe conditions, equipment issues, or environmental hazards are detected.
- **Occupancy and Usage Updates**
  The system updates tracked occupancy totals, visit records, and machine usage information so that staff and management can review operational trends over time.
- **Web Application Responses**
  The system returns information to the web application such as class registration confirmation, class availability status, membership-related updates, and current gym occupancy data.
- **Bulky Buzzer Activation**
  When disruptive noise conditions are detected or when staff manually activate the feature, the system sends a signal to the Bulky Buzzer to produce an alert in the physical gym space.

- **System Logging and Record Updates**
  The system stores relevant activity, alert, and membership-related data in the appropriate data stores for later review, auditing, and business analysis.

3.2.3 Relation to Physical Devices

The external interfaces of the GSMS are directly tied to the physical devices distributed throughout the gym. Entry and exit events originate from the front-door scanner, machine activity events come from equipped exercise machines, and environmental hazard signals come from thermostats and smoke/CO detectors placed in relevant areas. Camera-based interfaces provide monitoring coverage across workout zones and walkways, while the Bulky Buzzer serves as a physical output device that communicates directly into the gym environment. Staff and manager computers serve as the primary monitoring endpoints for alerts and system status updates, and the web application acts as the member-facing interface for class scheduling, gym access support, and occupancy review. Together, these devices form the bridge between the software and the real-world gym space.

3.3 Control Logic

In this section, we describe the control logic of the Gym Space Monitoring System (GSMS) using a dynamic model (state machine) based on the Object Modeling Technique (OMT). This model captures the system's primary operational states and the event-driven transitions that occur as sensor inputs, user interactions, and scheduled processes are received and processed. The following diagrams illustrate how events propagate through system components—such as monitoring modules, AI analysis, and management services—to generate alerts, update system data, and support real-time decision-making across both user-facing and administrative interfaces.

3.3.1 Alert Manager

The Alert Manager dynamic diagram in *Figure 2* illustrates the system's state-based control flow and shows how event-driven inputs are processed and managed throughout the alert lifecycle. The system initializes in an idle state, where it continuously monitors incoming events generated by system components such as sensors and AI analysis modules. Upon receiving an event, the system transitions to a processing state, where it determines the type and severity of the event to decide the appropriate response. The alert can be specified as an informational alert, where no action is required but the action is still logged. It can also be a warning alert, which is the standard alert which requires a staff member to validate and investigate the issue and make the judgement on how to resolve the situation. Then there are the critical alerts which require immediate validation and action from the staff.

Once classified, the system enters a dispatching state, where the alert is forwarded to the appropriate staff and manager dashboards for visibility and action. The right alert is sent, a timer is started, and a log is sent to the log store. After dispatch, the system transitions to a waiting state, where it awaits acknowledgment from staff. If acknowledgment is received within an acceptable timeframe, the system proceeds to a closing state, where the incident is logged, relevant system data is updated, and the alert is marked as resolved before returning to the idle state.

Alternatively, if no acknowledgment is received, the system transitions to an escalation state, where the alert is re-sent or elevated to ensure visibility. The system then returns to the waiting state and continues this cycle until acknowledgment is received. This behavior ensures that critical events are not overlooked and that all alerts are properly handled before the system resumes monitoring.

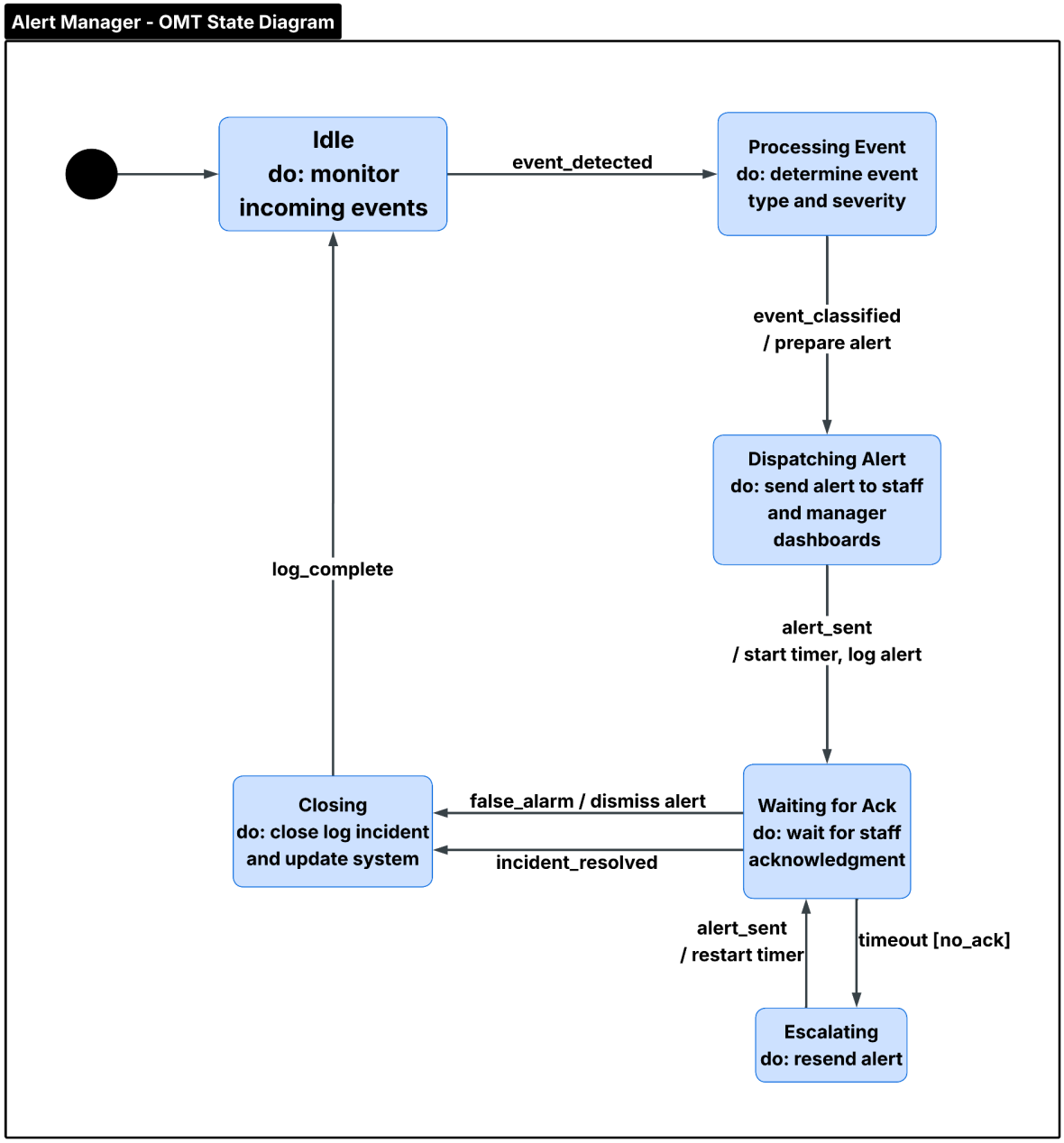

Figure 2: Alert Manager - OMT State Diagram

3.3.2 Door Scanner Controller

The Door Scanner Controller dynamic diagram in *Figure 3* illustrates the system's state-based control flow and shows how user scan events are processed to manage facility access. The system initializes in an idle state, where it waits for a QR scan input from a member. Upon receiving a scan, the system transitions to a reading state, where it captures the scanner input and proceeds to verify the member by requesting validation from the Membership Store.

If the membership is invalid, the system transitions to an access denied state, where a denial message is displayed before returning to the reset state and ultimately back to idle. If the membership is valid, the system proceeds to determine the context of the scan by entering a checking door type state, where it identifies the scanner location and whether additional validation is required. For restricted areas such as class-specific rooms, the system transitions to a class registration check, where it verifies whether the member is registered for the current or upcoming session. If the member is not authorized, access is denied and the system resets.

If all validation checks are satisfied, the system transitions to an unlocking state, where it sends a signal to unlock the door. Following successful entry, the system updates entry statistics, including occupancy and activity tracking, before transitioning to a reset state that clears the current scan session. The system then returns to the idle state to await the next scan event. This process ensures that access control is both secure and context-aware while maintaining accurate system data.

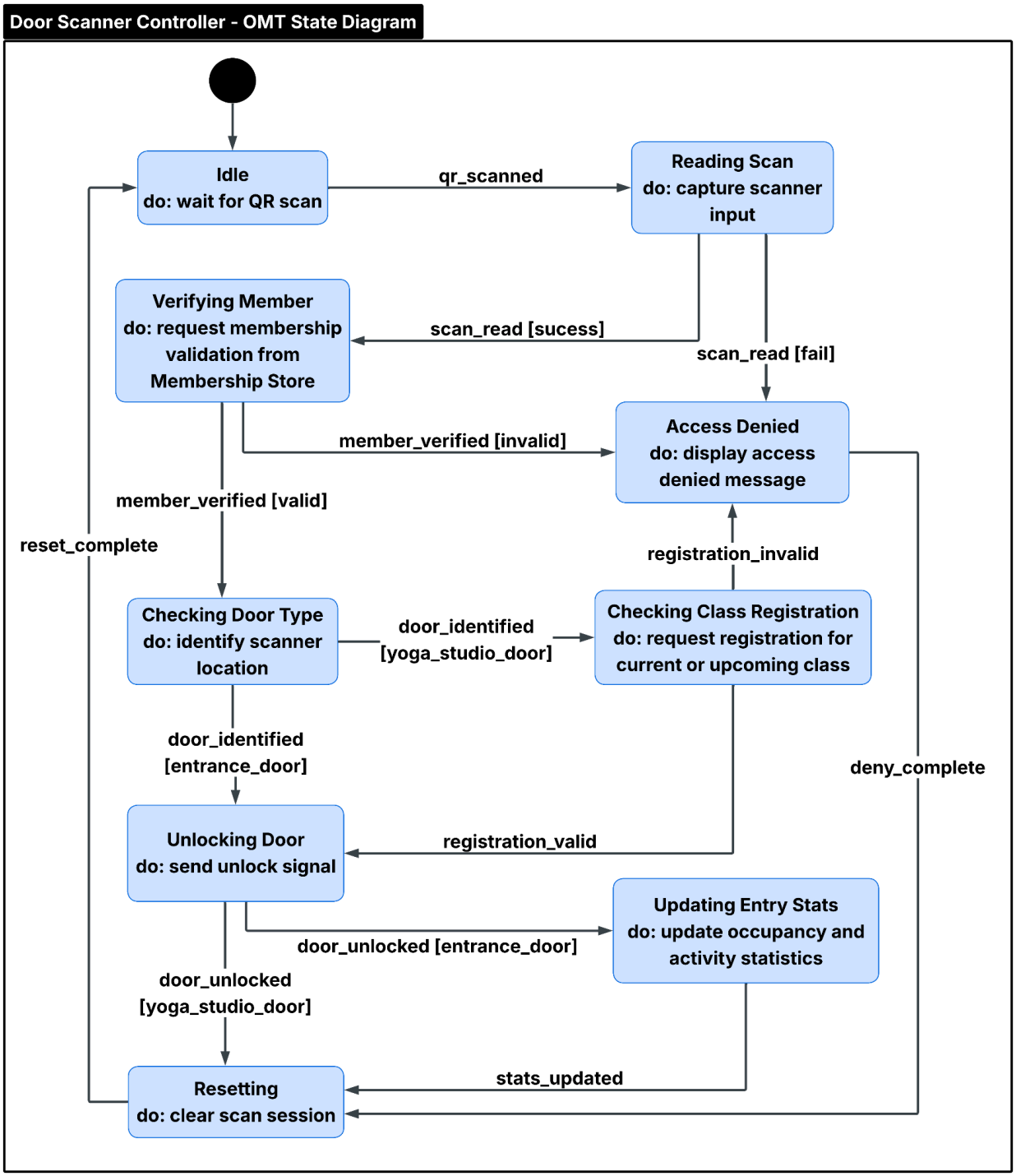

Figure 3: Door Scanner Controller - OMT State Diagram

3.3.3 Employee Dashboard

The Employee Dashboard dynamic diagram in *Figure 4* illustrates the system's state-based control flow and shows how user interactions are processed to monitor, view, and respond to alerts and system data. The system initializes in a sign-in state, where it displays the login interface and waits for user credentials. Upon submission, the system transitions to a verifying state, where the employee ID is validated and the user's role is determined.

Once authenticated, the system transitions to the appropriate dashboard state based on user role. Standard employees are directed to the employee dashboard, where active alerts are displayed, while managers are directed to the manager dashboard, which provides access to alerts, accessing system logs from the log store, and gym activity data. From these states, users may transition to viewing alerts, viewing logs, or viewing statistics, depending on the selected action.

When an alert is selected, the system transitions to a viewing alerts state, where alert details are displayed. If the user chooses to take action, the system transitions to a verification state to ensure the user is authorized. Upon successful verification, the system enters a handling alert state, where alert details and response options are presented. For actions requiring elevated privileges, the system transitions to a manager override state, where a manager PIN is required before proceeding. Managers may also directly transition to a manager handling state, where they are provided with extended resolution options.

After completing an action or navigating between views, the system returns to the appropriate dashboard state and continues operation. This behavior ensures consistent, role-based access control while supporting efficient monitoring and response to system events.

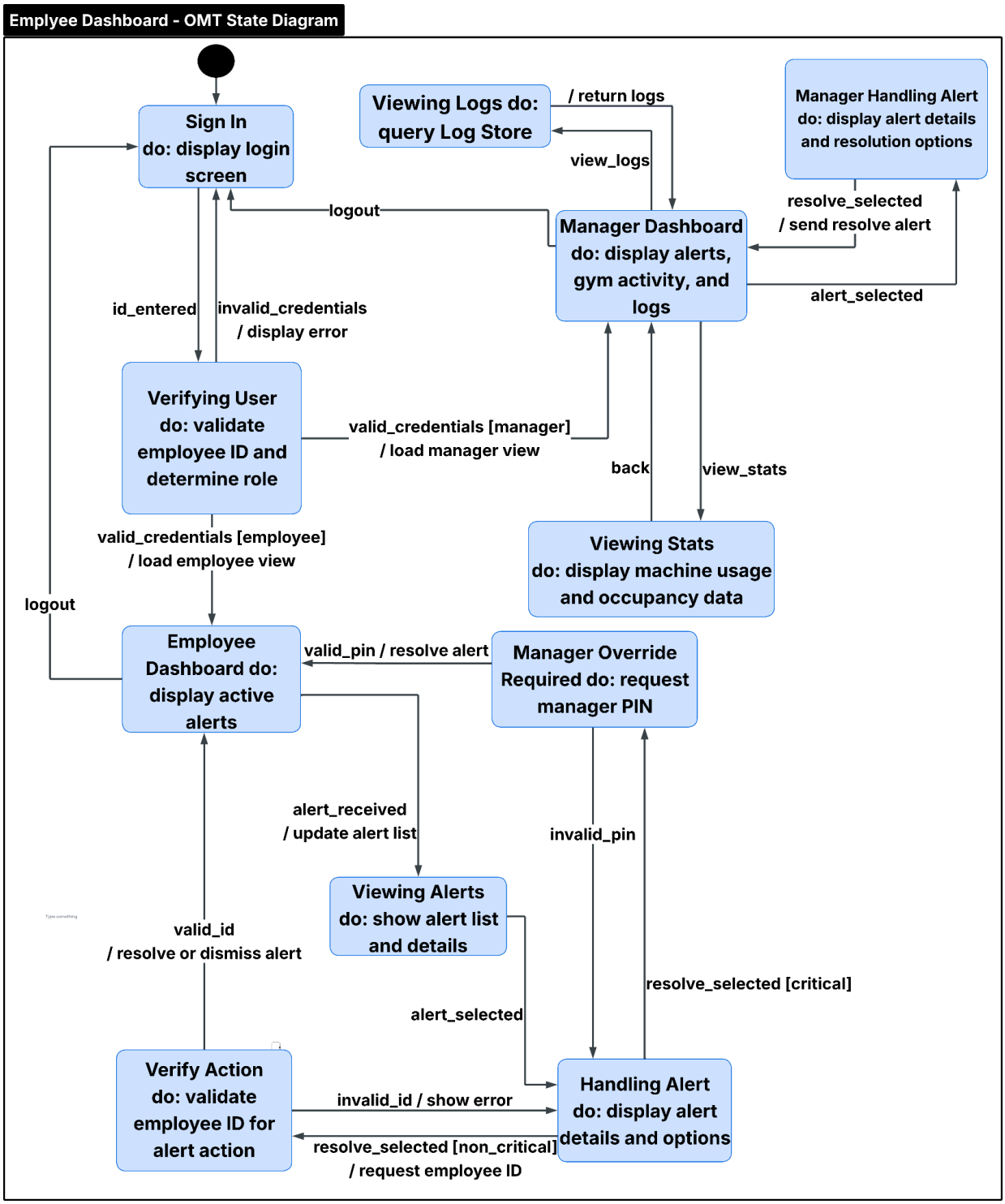

Figure 4: Employee Dashboard - OMT State Diagram

3.3.4 Web Service

The Web Service dynamic diagram in *Figure 5* illustrates the system's state-based control flow and shows how user interactions are processed to manage authentication, account access, and class-related functionality. The system initializes in an idle state, where it displays the sign-in interface and authentication options. From this state, users may either submit login credentials or begin the registration process.

Upon receiving login input, the system transitions to a verifying credentials state, where the username and password are validated. If the credentials are invalid, the system transitions to an error state, where an appropriate message is displayed before returning to the sign-in state. Similarly, during the registration process, the system transitions to a processing sign-up state, where new member information is collected. If an error occurs during this process, the system transitions to the error state before returning to the sign-in state.

If authentication or registration is successful, the system transitions to the home page state, where class information and account options are displayed. From this state, users may initiate additional actions such as registering for a class, viewing their QR access code, or logging out. When registering for a class, the system transitions to a processing class registration state, where the request is submitted and then returns to the home page. When viewing access credentials, the system transitions to a displaying QR code state, where the member's access code is presented before returning to the home page.

If the user chooses to log out, the system transitions to a logging out state, where the session is terminated before returning to the sign-in state. This behavior ensures secure authentication, consistent user navigation, and reliable interaction with membership and class management features.

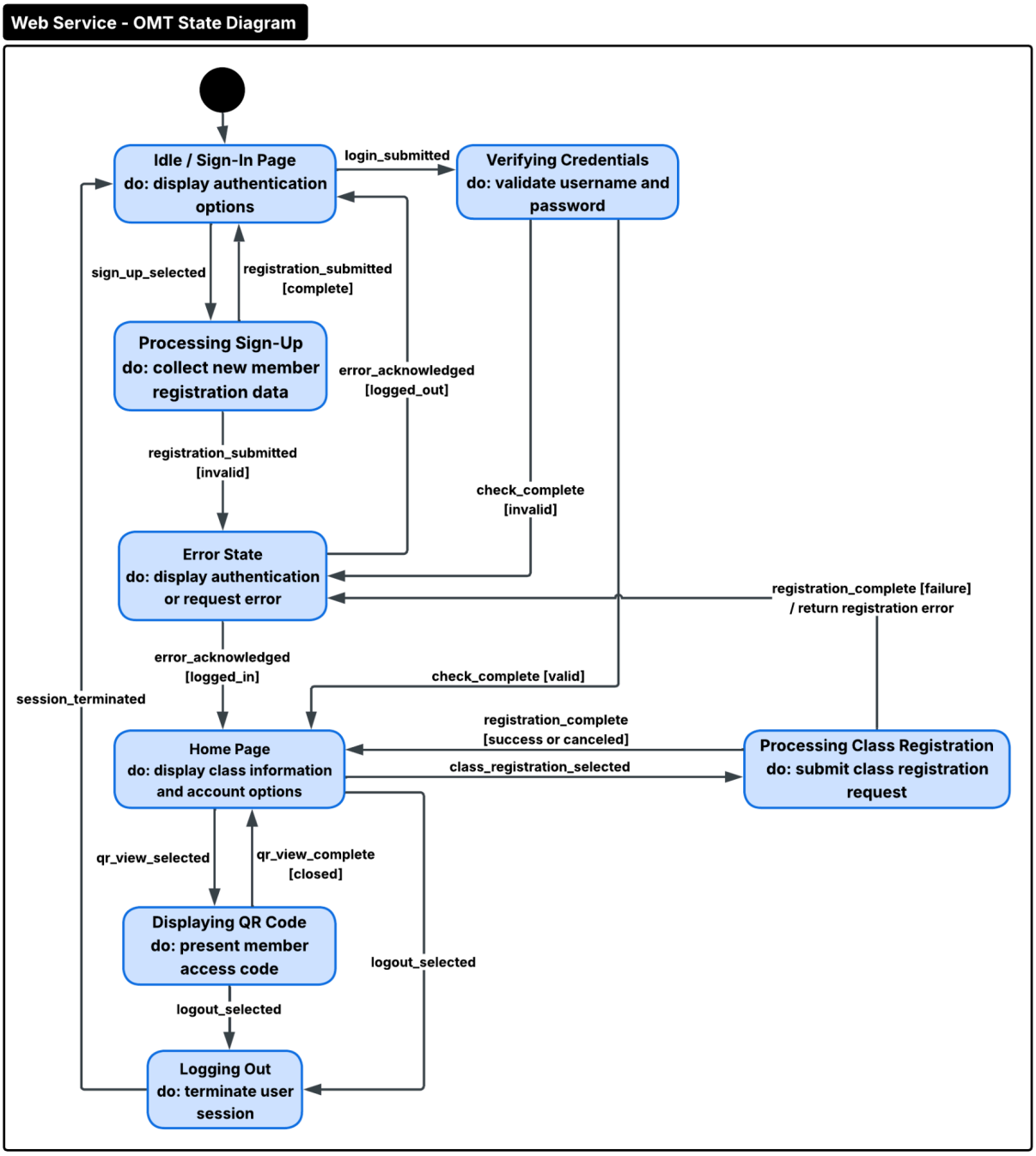

Figure 5: Web Service - OMT State Diagram

3.3.6 AI Hazard Analyzer

The AI Hazard Analyzer dynamic diagram in *Figure 6* illustrates the system's state-based control flow and shows how visual input is processed to detect safety-related events within the gym environment. The system initializes in an idle state, where it awaits incoming camera frames from the monitoring system. Upon receiving a frame, the system transitions to an analyzing frame state, where the input is evaluated for potential hazards, unsafe behavior, or abnormal conditions, through api calls to a computer vision AI.

If no hazard is detected, the system returns to the idle state and continues monitoring incoming frames. If a hazard or safety event is identified, the system transitions to a reporting detection state, where the detected event data is sent to the Alert Manager for further processing and response. After reporting the event, the system returns to the idle state to continue continuous monitoring.

This behavior ensures real-time analysis of the gym environment while maintaining a consistent event-driven flow that integrates with the broader alert management system.

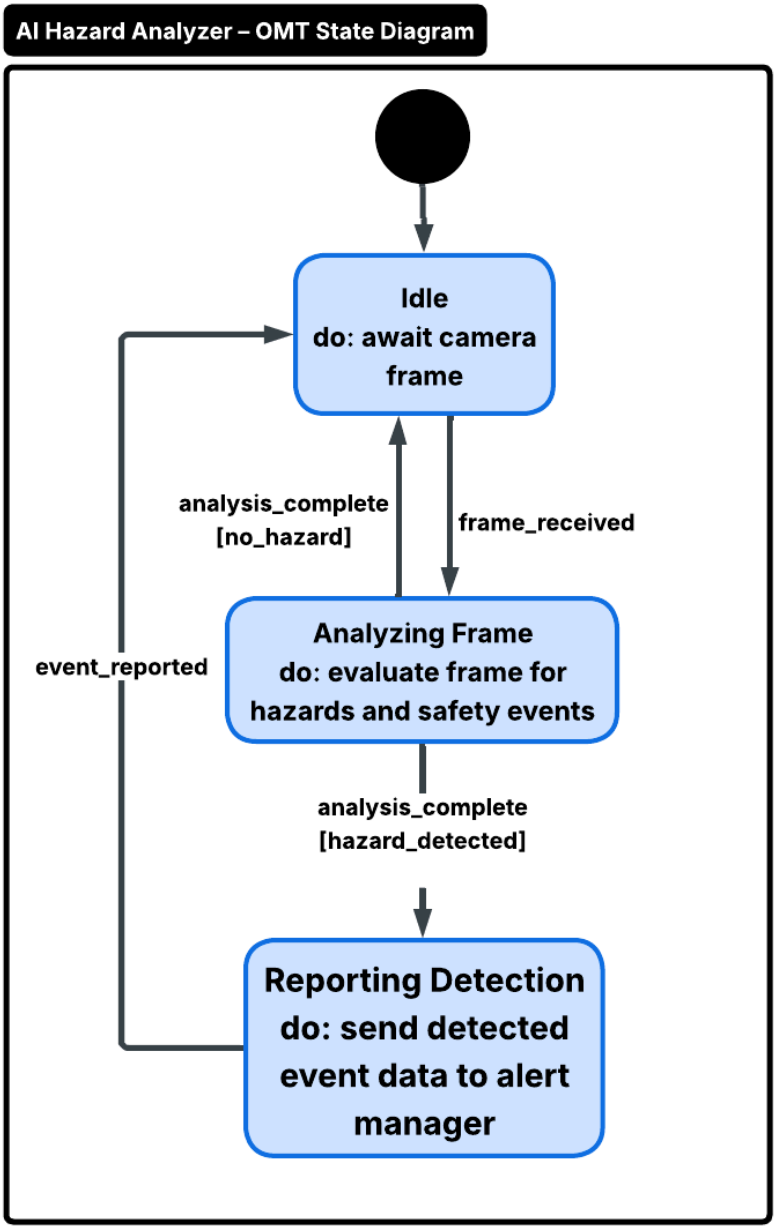

Figure 6: Subscription Manager - OMT State Diagram

3.3.7 Sound Monitor

The Sound Monitor dynamic diagram in *Figure 7* illustrates the system's state-based control flow and shows how ambient noise levels are monitored to detect unsafe or disruptive conditions. The system initializes in an idle state, where it continuously monitors ambient sound levels within the gym environment.

When the detected sound level exceeds a predefined threshold, the system transitions to an activating buzzer state, where a buzzer alert is triggered to notify occupants and encourage corrective behavior. Once the sound level returns to an acceptable range, the system transitions back to the idle state and resumes continuous monitoring.

This behavior ensures that excessive noise levels are promptly addressed while maintaining a simple and responsive monitoring loop.

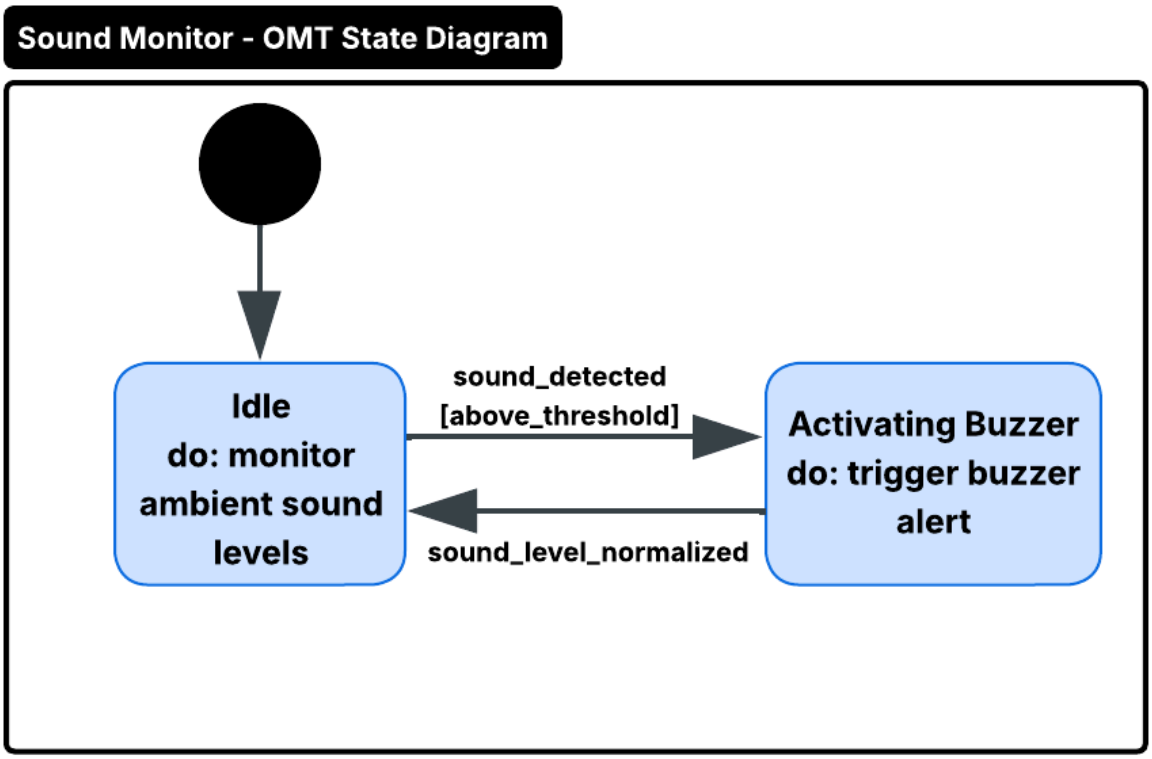

Figure 7: Sound Monitor - OMT State Diagram

3.3.8 Air Quality/Temp Monitor

The Air Quality/Temperature Monitor dynamic diagram in *Figure 8* illustrates the system's state-based control flow and shows how environmental conditions are monitored to detect

unsafe temperature or air quality levels. The system initializes in an idle state, where it continuously monitors temperature and air quality within the gym environment.

When the detected conditions exceed predefined safety thresholds, the system transitions to a sending critical alert state, where an environmental hazard alert is dispatched to the Alert Manager for further processing and response. Once conditions return to acceptable levels, the system transitions back to the idle state and resumes continuous monitoring.

This behavior ensures that environmental hazards are detected and reported in a timely manner while maintaining a consistent and responsive monitoring loop.

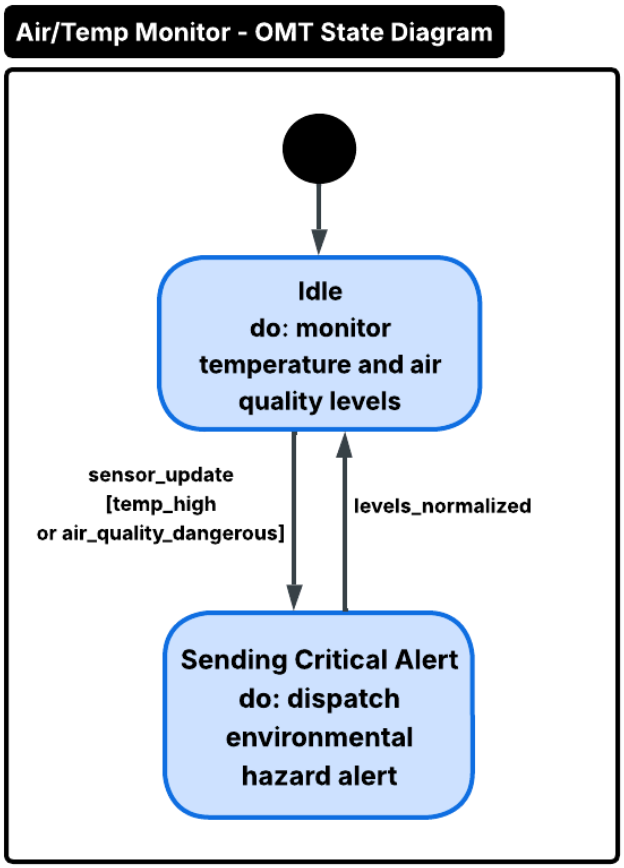

Figure 8: Air Quality/Temp Monitor

3.3.9 Machine Monitor

The Gym Machine Monitor dynamic diagram in *Figure 9* illustrates the system's state-based control flow and shows how equipment usage and operational status are monitored to detect faults or abnormal behavior. The system initializes in an off state, where it monitors

whether a machine is idle. When activity is detected, the system transitions to an in use state, where it monitors active machine operation and usage patterns.

If the machine returns to an idle condition, the system transitions back to the off state and continues monitoring. If a fault or abnormal condition is detected during operation, the system transitions to a fault detected state, where an equipment fault alert is generated and sent to the Alert Manager for further processing. After issuing the alert, the system transitions back to the off state and resumes monitoring.

This behavior ensures continuous tracking of equipment usage while enabling timely detection and reporting of machine faults.

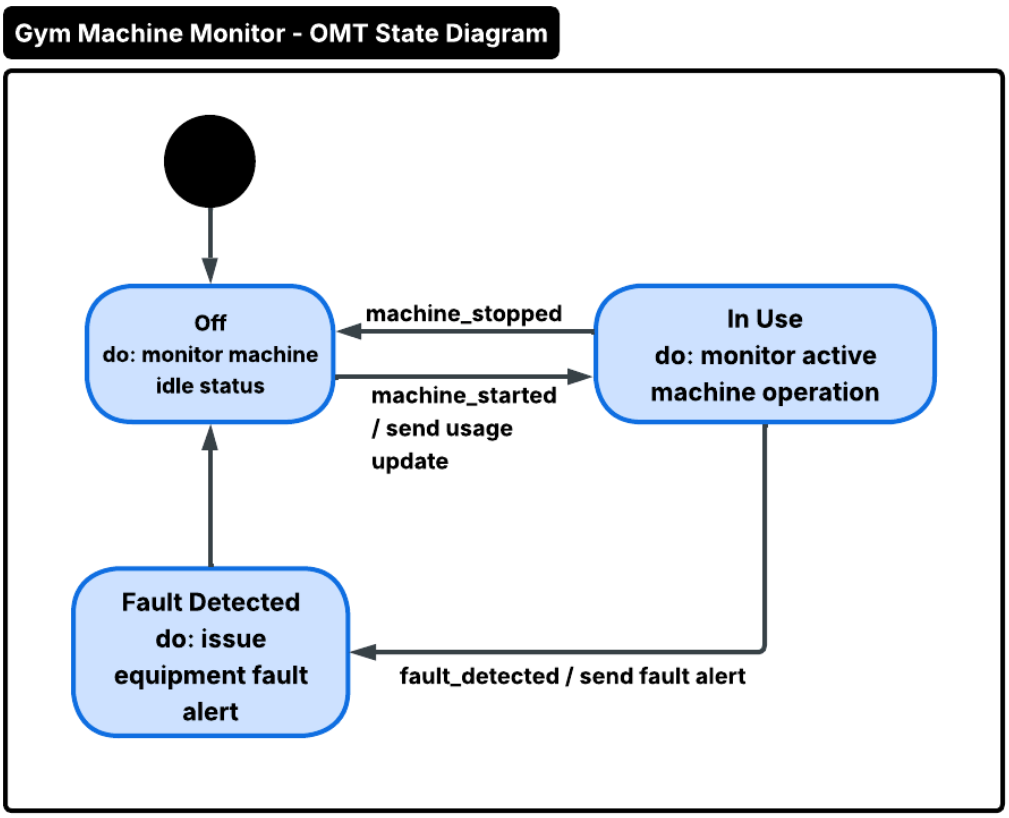

Figure 9: Gym Machine Monitor

4 Design Constraints

The design, which was laid out, was built with several constraints in mind. These are specifications and limitations set forth by the client, the law, the gym environment, equipment

limitations, and the use of AI-based analysis. Within these guidelines, the system will operate in a manner that is reliable, operational, and easy to manage.

4.1 Maintainability

The system is designed to be maintainable for both hardware and software components. Since the system is deployed across multiple gym zones with distributed sensors and cameras, failures must be easy to identify and resolve.

To support maintainability, the system incorporates:

- Error logging and alert mechanisms for sensor failures and AI missclassifications
- Modular hardware components (e.g., cameras and sensors) that can easily be replaced without disrupting the system
- A centralized monitoring interface that allows staff to quickly identify malfunctioning equipment or system components

On the software side, the system leverages modular design and external AI APIs, allowing updates or improvements to be made without requiring a full system redesign. This ensures long-term adaptability and easier maintenance as system requirements evolve.

4.2 AI and Data Processing Constraints

The GSMS relies heavily on AI-based analysis of camera data to detect events such as falls, overcrowding, and unsafe behavior. However, AI introduces limitations in accuracy, latency, and contextual understanding.

To address these constraints, the system is designed to:

- Use external AI API services instead of locally hosted models, reducing computational overhead
- Treat AI outputs as advisory rather than authoritative, requiring conservative decision-making
- Handle uncertainty by limiting high-risk automated actions and prioritizing staff alerts

These constraints directly influence the system's architecture, requiring integration between real-time data streams, AI outputs, and alerting mechanisms.

4.3 Environmental Constraints

The gym environment presents challenges such as humidity, heat, sweat exposure, and constant physical activity. These conditions can impact both hardware reliability and data quality.

To address these constraints, the system is designed such that:

- Hardware components are selected to withstand moisture and temperature variations
- Camera placement and sensor positioning are optimized to reduce occlusion and interference
- Software accounts for noisy or inconsistent data caused by environmental conditions

These considerations ensure that the system continues to operate reliably in a high-traffic, physically dynamic environment.

4.4 Privacy Constraints

Because the system uses camera-based monitoring, privacy is a critical design constraint. As outlined in the system organization, certain areas, such as locker rooms, are treated as privacy-sensitive zones and exclude camera monitoring.

To address these constraints, the system is design:

- Restricts camera usage in sensitive areas
- Relies on non-visual sensors (e.g., environmental sensors) where appropriate
- Limits stored data to necessary operational and safety information

These constraints ensure that the system balances safety monitoring with user privacy expectations.

4.5 Equipment and Integration Constraints

The GSMS must operate without interfering with the functionality of existing gym equipment. As described in the system organization, the gym includes both sensor-enabled machines and non-instrumented equipment, such as free weights

To address these constraints, the system is designed to:

- Uses non-invasive monitoring methods, such as external sensors and camera-based observation
- Avoids direct integration with internal machine electronics

- Relies on AI-based visual analysis in areas where direct sensing is not possible (e.g., free weights area)

Additionally, the system must account for equipment malfunctions. If a machine is detected to be unsafe or producing inconsistent data, the system flags it and avoids generating recommendations or alerts related to its use.

4.6 System Reliability and Recovery

The GSMS must provide continuous monitoring and timely alerts, even in the presence of partial system failures or disruptions.

To support reliability, the system is designed to:

- Degrade gracefully when components fail (e.g., disabling features tied to failed sensors)
- Maintain consistent communication between distributed components and the central computer
- Recover quickly after power loss or system restart, restoring monitoring and alerting functionality without requiring manual reconfiguration

This constraint influences both the system architecture and the choice of components, ensuring consistent performance in real-world conditions.

5 Definition of Terms

Throughout the document, certain words will be used frequently when describing the GSMS. The following words are defined precisely in how we intend to use them, and what they refer to when it comes to the GSMS in particular.

- **Gym Space Monitoring System (GSMS):** The official title refers to the whole system, which includes all devices in each room, along with the software components.

- **Management:** The overall administrative acts used to manage the gym, including member sign-up and tracking, upholding building and fire codes, member safety, as well as managing the traffic and flow of members within the gym.

- **Central Computer:** The primary processing unit of the GSMS is responsible for aggregating data from all sensors, performing analysis, and coordinating alerts and system responses.

- **Maintenance:** Referring to the upkeep of the gym as it relates to gym equipment. This refers to the replacement and fixing of faulty or broken gym equipment, along with hygienic practices such as cleaning all equipment at regular intervals.

- **Identification Scanner:** A scanner which is placed at the entrance of the gym, where members are provided with an ID card upon sign-up and can then scan in and scan out of the gym using simple 2D barcode scanning.

- **Bulky Buzzer:** A special alarm which is placed near the front of the gym, which triggers if people or a particular member is being too loud during their exercise, thus making other members uncomfortable and unsafe.

- **Environmental Hazards:** Refers to both extreme hazards, such as fires and other emergencies that warrant evacuation, and to small hazards such as tripping on misplaced gym equipment.

- **Camera:** An AI-powered camera that uses computer vision techniques to monitor gym activity and identify predefined conditions. These cameras are designed to detect indicators of unsafe behavior (e.g., improper form or potential falls) and estimate occupancy within their field of view, enabling the system

- **Alerts:** Messages which are sent to various components throughout the GSMS. There are 3 types of alerts, an informational alert is a type of alert which strictly signals a longing event. A warning alert which is both logged and displayed on employee dashboards, prompting staff to investigate. Then finally a critical alert which is the highest priority alert, it is logged and prompts staff to look into and resolve immediately.

- **Log Store:** The datastore where the system both reads and writes log information.

6 References

[1] Lucid Software Inc. "Lucidchart." Lucid Software, www.lucidchart.com.
[2] *Fire Code*, www.cabq.gov/fire/documents/city-of-albuquerque-fire-code.pdf. Accessed 28 Mar. 2026.
[3] "NEW MEXICO COMMERCIAL BUILDING CODE." *14.7.2 NMAC*, www.srca.nm.gov/parts/title14/14.007.0002.html. Accessed 28 Mar. 2026.